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Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 17.09.2025

Mobile Applications for IOS

Introduction

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Consultations

Consultations (Microsoft Teams / C-3, s.122) - please make an appointment for a consultation by e-mail:

- Tue. 15:00 - 17:00 - Microsoft Teams
 - Thu. 16:30 - 18:30 - C-3, s.122
-



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Presence

Attendance will be checked during three selected lectures.



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Passing the lecture

Attendance (2/3): Attendance will be checked during three selected lectures. Being present at least twice guarantees that the laboratory grade will count toward the lecture grade.

or

Written exam. The dates will be determined in subsequent lectures.



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Coding for iOS Devices

Tools

- **Xcode** (macOS) – full-featured IDE for building, testing, and publishing iOS apps;
- **Swift Playgrounds** (macOS, iPadOS) – interactive environment for learning and experimenting with Swift.

Languages

- **Swift** – main programming language for iOS;
- **SwiftUI** – modern framework for building declarative user interfaces

Getting Started

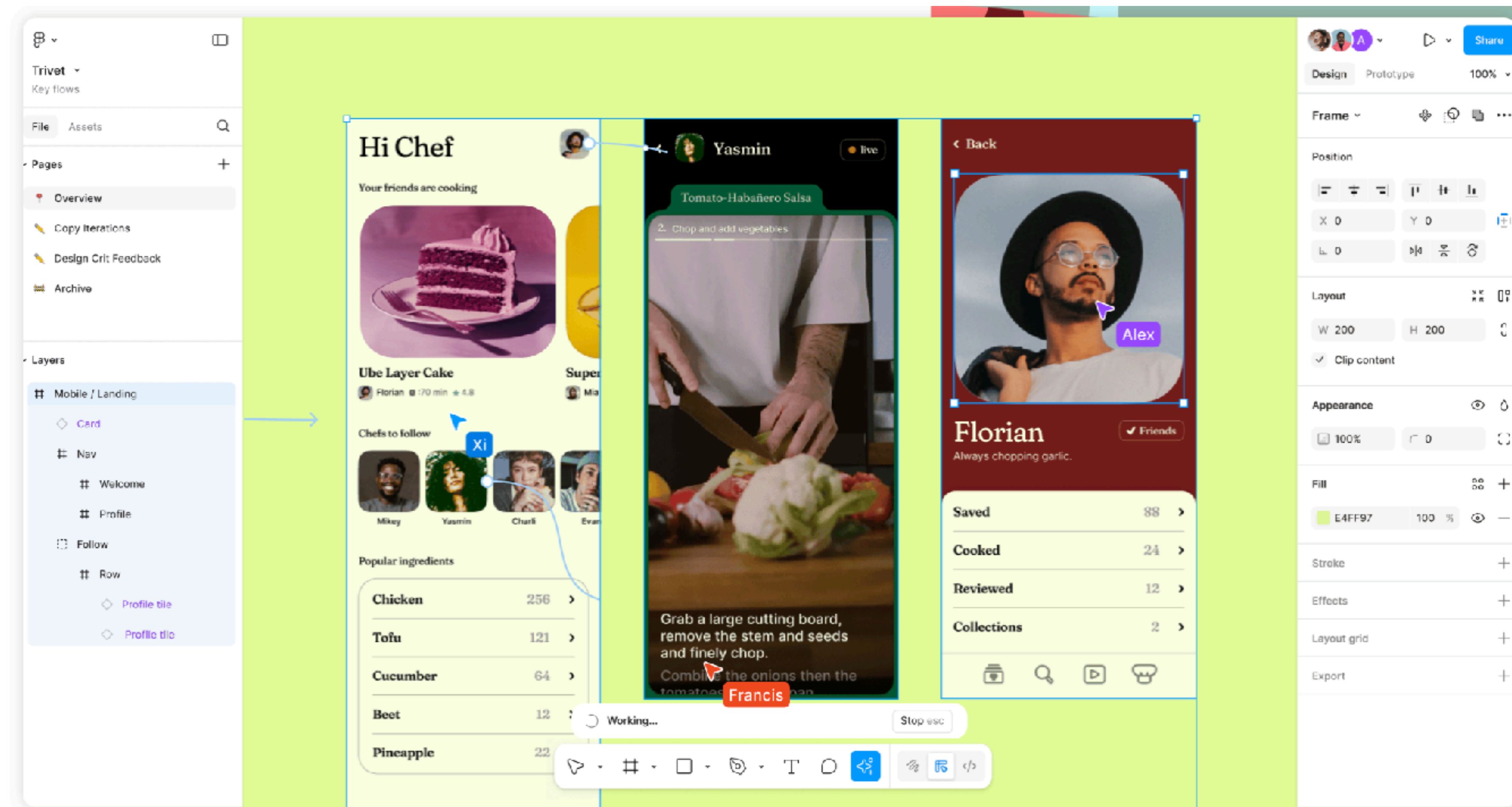
- Begin with Swift Playgrounds to learn the basics
- Move to Xcode for full app development and App Store deployment



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Figma





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Figma



Apple

iOS and iPadOS 26

Design file • ♥ 13.9k • 👤 101k users

Open in Figma

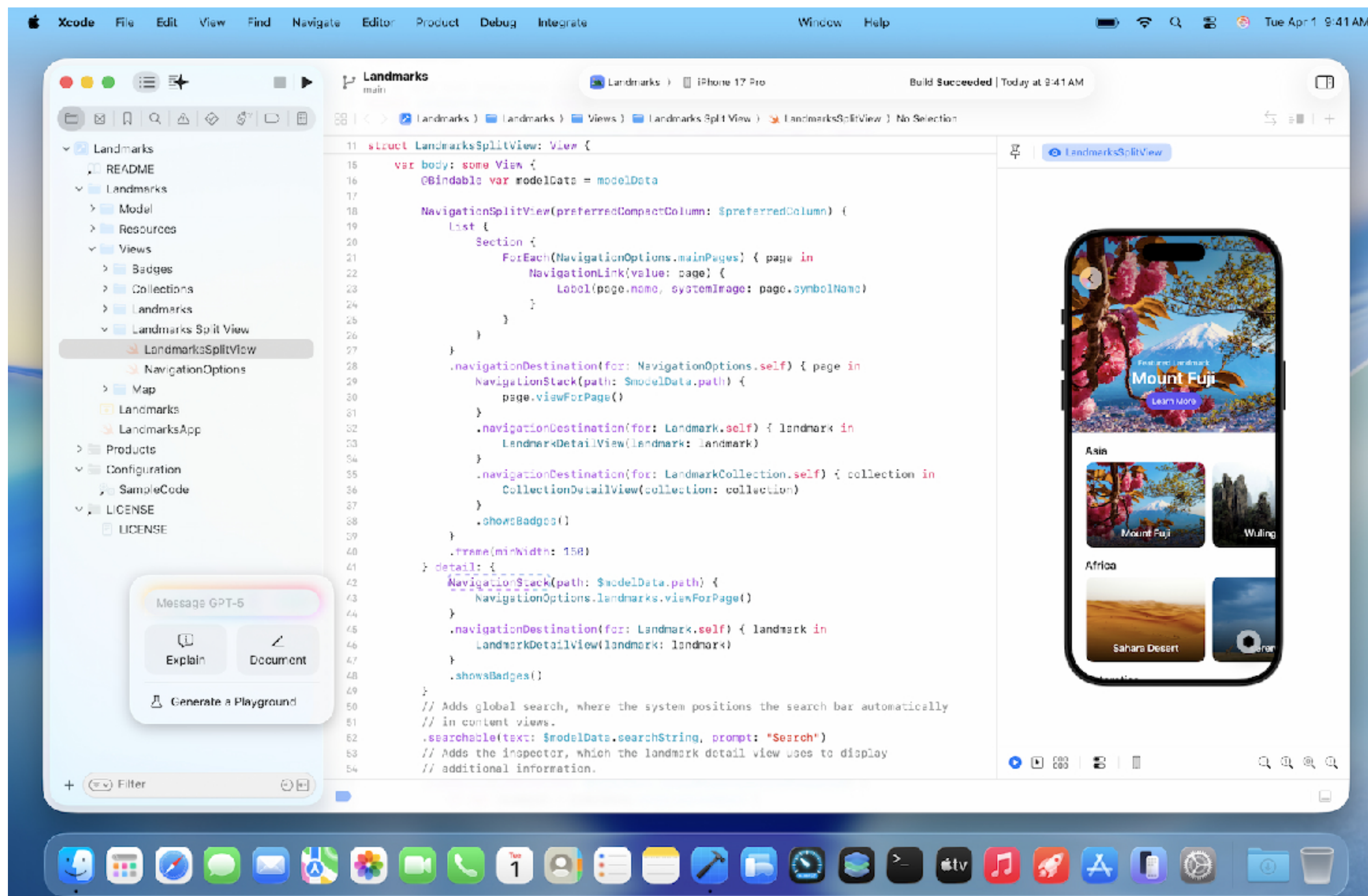




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Xcode



Source: <https://developer.apple.com/xcode/>



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Xcode



Xcode offers the tools you need to develop, test, and distribute apps for Apple platforms, including predictive code completion, generative intelligence powered by the best coding models, advanced profiling and debugging tools, and simulators for Apple devices.

macOS Sequoia 15.7.1

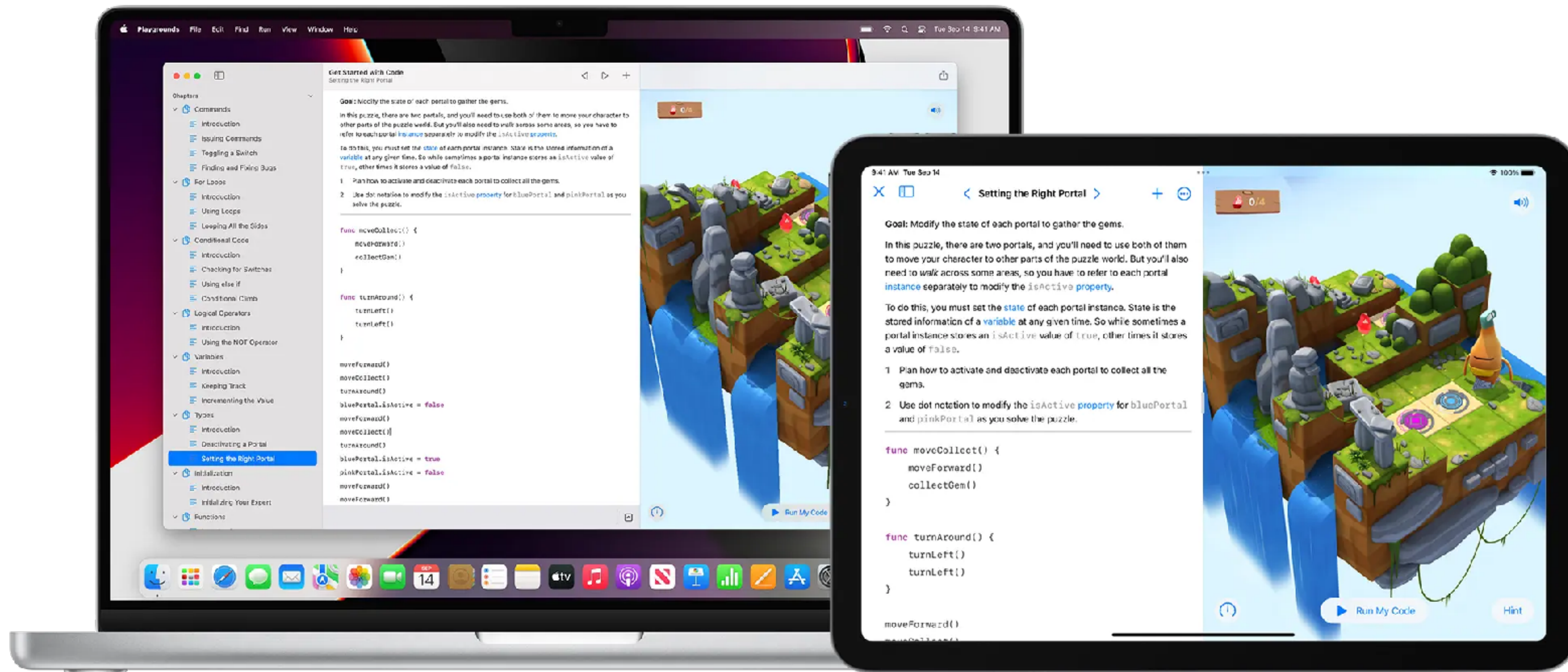
macOS Tahoe 26.0.1



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Swift Playground





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Swift Playground



Swift Playground is a revolutionary app for iPad and Mac that helps you learn to code and build apps using Swift, the same powerful language used to create world-class apps for the App Store. Engaging lessons and walkthroughs demonstrate the core concepts of coding and building apps as you write real Swift code in an interactive environment.

iPadOS 18

iPadOS 26.0.1



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Lecture schedule



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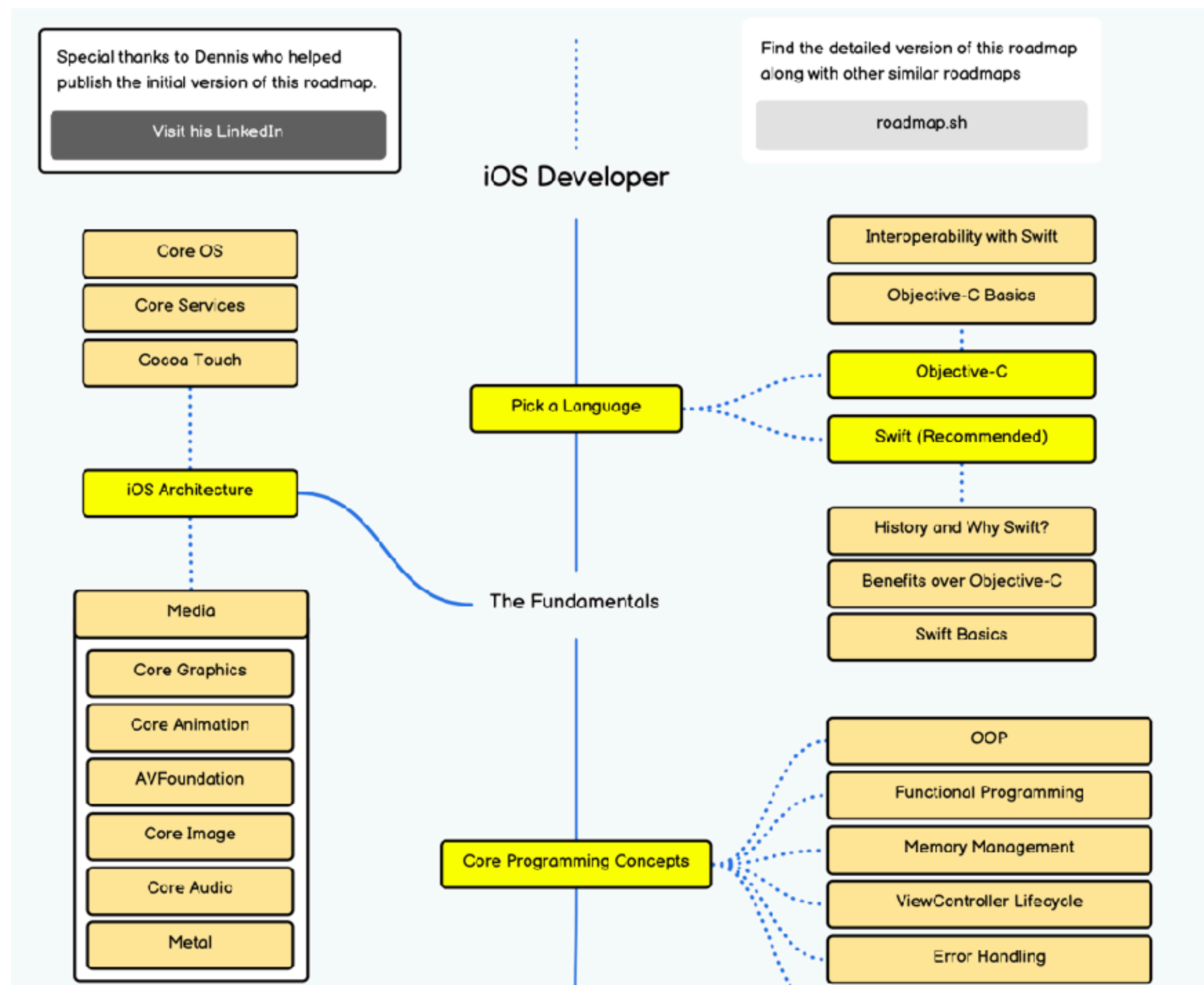
PROGRAMME CONTENT		
Lecture		Number of hours
Lec 1	Introduction. Review of selected mobile applications. Introduction to Apple devices and Mac OS.	2
Lec 2-3	UI and UX. Interactions. Touch screens. Material design. Human Interface guidelines. Voice communication. Siri.	4
Lec 4	Tools for multimedia processing. Gathering multimedia data. Copyright.	2
Lec 5	Implementation rules in Xcode. The structure of application written in Swift.	2
Lec 6-7	Presentation of basic elements of Swift by examples. Core libraries.	4
Lec 8	Apple Human Interface Guideline by examples.	2
Lec 9	Libraries and frameworks supporting creation of multimedia applications. Short characteristics of Kotlin language.	2
Lec 10	Applications of augmented reality. Code analysis of application with augmented reality mechanisms.	2
Lec 11	Games review. Connections between games and the progress of algorithms and programming languages.	2
Lec 12	Multimedia data compression. Compression formats. Video and audio streaming.	2
Lec 13	Multimedia in mobile systems. Cameras. Recommended frameworks.	2
Lec 14	Interaction mechanisms – review. New technologies and multimedia devices.	2
Lec 15	Summary. Perspectives of multimedia techniques.	2
	Total hours	30



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Roadmap





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Useful links

- **Apple Developer:** <https://developer.apple.com/>
- **Swift.org:** <https://www.swift.org/>
- **iOS Human Interface Guidelines:** <https://developer.apple.com/design/human-interface-guidelines/designing-for-ios>
- **Hacking with Swift:** <https://www.hackingwithswift.com/>
- **iOS App Distribution Guide:** <https://developer.apple.com/documentation/appstoreconnectapi>
- **iOS App Dev Tutorial:** <https://developer.apple.com/tutorials/app-dev-training/getting-started-with-today>

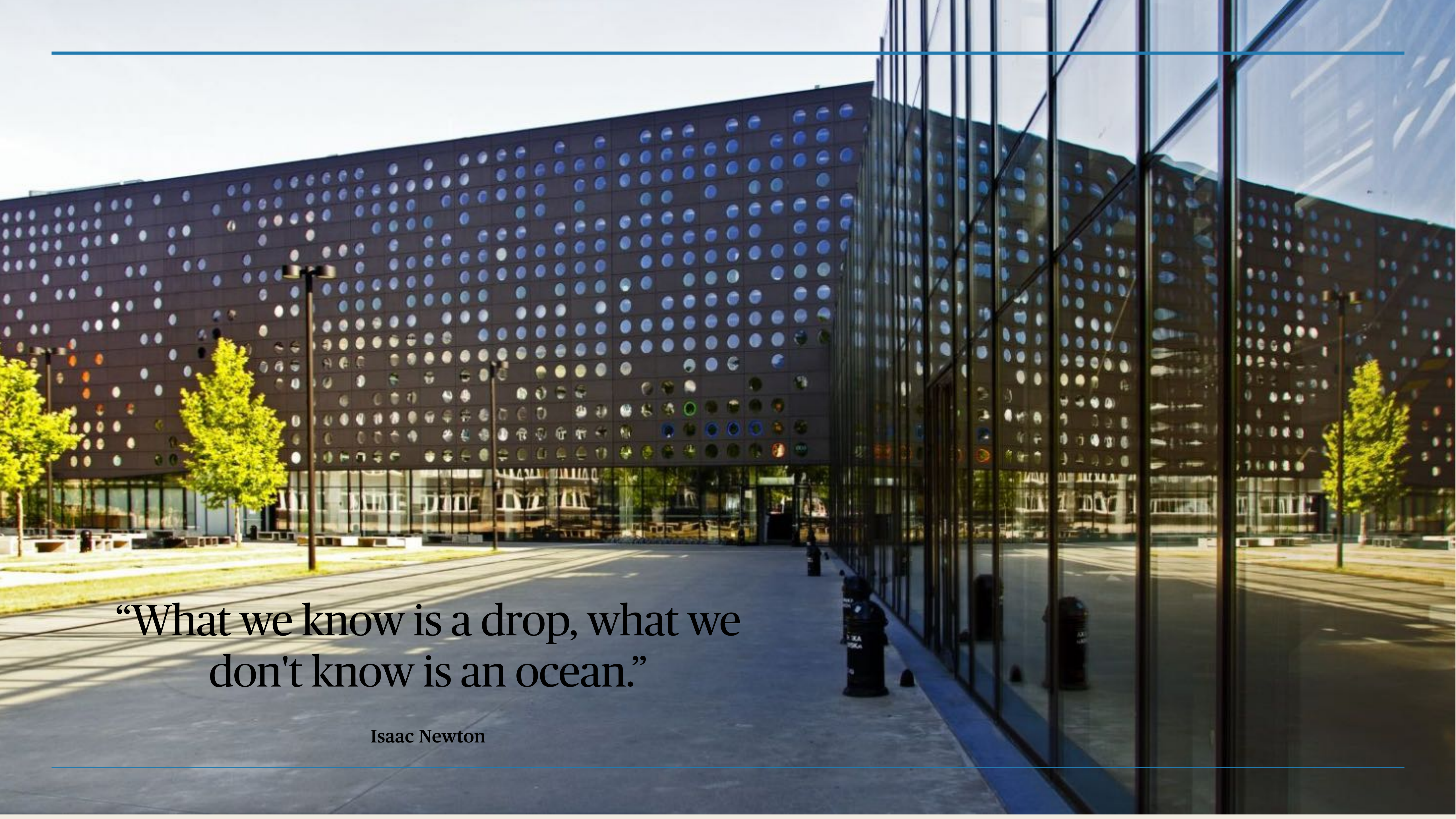


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Books





“What we know is a drop, what we
don't know is an ocean.”

Isaac Newton
